Self-consistent Hartree-Fock--Bogoliubov theory

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I. HARTREE-FOCK-BOGOLIUBOV MEAN-FIELD THEORY

A. Introduction

Consider an interacting many-body system of electrons described by the Hamiltonian

$$H = H_0 + H_{\text{int}} = \sum_{\alpha\beta} h_{\alpha\beta} c^\dagger_\alpha c_\beta + \sum_{\alpha\beta\gamma\delta} V^{\alpha\beta}_{\gamma\delta} c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta, \tag{1}$$

with $c^\dagger(c)$ fermionic creation (annihilation) operators and the Greek indices α, β, γ, δ generic degrees of freedom. The time-dependency on the fermionic operators are being omitted for clarity, as is also done in the non-interacting Hamiltonian, accounting for a coupling to a perturbative external force, i.e $h_{\alpha\beta} \to h_{\alpha\beta} + B_{\alpha\beta}F(t)$. Concerning the interaction term, we assume $V(\mathbf{r})$ to be generic electrostatic potential of symmetry $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$ such that the interacting potential tensor has the symmetries $V^{\alpha\beta}_{\gamma\delta} = V^{\beta\alpha}_{\delta\gamma}$ and $(V^{\alpha\beta}_{\gamma\delta})^* = V^{\gamma\delta}_{\alpha\beta}$.

Symmetry properties of V
From the imposed symmetry $V(\mathbf{r}-\mathbf{r}') = V(\mathbf{r}'-\mathbf{r})$, we obtain

$$\begin{aligned} V^{\alpha\beta}_{\gamma\delta} &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\alpha(\mathbf{r})\phi^*_\beta(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r}')\phi_\delta(\mathbf{r}),\ \mathbf{r} \leftrightharpoons \mathbf{r}' \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\alpha(\mathbf{r}')\phi^*_\beta(\mathbf{r})V(\mathbf{r}-\mathbf{r}')\phi_\gamma(\mathbf{r})\phi_\delta(\mathbf{r}') \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\beta(\mathbf{r})\phi^*_\alpha(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\delta(\mathbf{r}')\phi_\gamma(\mathbf{r}) = V^{\beta\alpha}_{\delta\gamma} \ \square \end{aligned}$$

Similarly, we can show that

$$\begin{aligned} \left(V^{\alpha\beta}_{\gamma\delta}\right)^* &= \int d\mathbf{r}d\mathbf{r}' \phi_\alpha(\mathbf{r})\phi_\beta(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi^*_\gamma(\mathbf{r}')\phi^*_\delta(\mathbf{r}) \\ &= \int d\mathbf{r}d\mathbf{r}' \phi^*_\delta(\mathbf{r})\phi^*_\gamma(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_\delta(\mathbf{r}')\phi_\alpha(\mathbf{r}) = V^{\delta\gamma}_{\beta\alpha} = V^{\gamma\delta}_{\alpha\beta} \ \square \end{aligned}$$

In order to study the dynamics of this system, we solve for the time-evolution of the the system's reduced density matrix (rDM) through it's its equation of motion (EoM). The system's purely electronic reduced density matrix (rDM) is defined as $\rho^{\text{ee}}_{ba} = \langle c^\dagger_a c_b \rangle$. However, if one wishes to study superconducting systems, one must also account for the anomalous terms. For this, we introduce the Nambu spinor representation with doubling of degrees of freedom $\check{c}_a = (c_a \ c^\dagger_a)$. The Nambu representation of the rDM, can then be written as the tensor product of this spinors for each of the spin combinations,

$$\rho_{ab} = \begin{pmatrix} \langle c^\dagger_b c_a \rangle & \langle c_b c_a \rangle \\ \langle c^\dagger_b c^\dagger_a \rangle & \langle c_b c^\dagger_a \rangle \end{pmatrix} = \begin{pmatrix} \rho^{\text{ee}}_{ab} & \rho^{\text{eh}}_{ab} \\ \rho^{\text{he}}_{ab} & \rho^{\text{hh}}_{ab} \end{pmatrix}. \tag{2}$$

Directly from the (equal-time) fermionic anti-commutator properties, this terms relate to each other as $\rho^{\text{hh}}_{ab} = -\rho^{\text{ee}}_{ba} + \delta_{ab}$ and $\rho^{\text{he}}_{ab} = (\rho^{\text{eh}}_{ba})^*$, but also $\rho^{\text{ee/hh}}_{ab} = (\rho^{\text{ee/hh}}_{ba})^*$ and $\rho^{\text{eh/he}}_{ba} = -\rho^{\text{eh/he}}_{ab}$. Notice that, since the spin degree of freedom was not yet specified we are not yet restricting ourselves to typical s-wave spin-singlet superconductors, this is a general treatment. Our task now is then to solve the two distinct EoMs, for the purely electronic rDM ρ^{ee}_{ab}, which then determines the other two simple through their relations.

B. Reduced density matrices equations of motion

We start by solving for the purely electronic rDM equation of motions

$$\frac{d}{dt}\rho^{\text{ee}}_{ab}(t) = \left\langle \frac{dc^\dagger_b(t)}{dt} c_a(t) \right\rangle + \left\langle c^\dagger_b(t) \frac{dc_a(t)}{dt} \right\rangle, \tag{3}$$

in the Heisenberg picture of quantum mechanics, where the fermionic operators evolve accordingly to the Heisenberg equation. Therefore the rDM equation of motion reads

$$\frac{d}{dt}\rho^{\mathrm{ee}}_{ab} = \left\langle \frac{i}{\hbar}[H, c_b^\dagger]c_a \right\rangle + \left\langle c_b^\dagger \frac{i}{\hbar}[H, c_a] \right\rangle . \tag{4}$$

where we have omitted the time dependency in the fermionic operators for compactness. This practice is done throughout the remaining work with no further mention. As shown in Appendix, given the equal-time fermionic anti-commutator properties $\left\{c_a, c_b^\dagger\right\} = \left\{c_b^\dagger, c_a\right\} = \delta_{ab}$ and $\{c_a, c_b\} = \left\{c_b^\dagger, c_a^\dagger\right\} = 0$, each of the commutators are evaluated to

$$[H, c_b^\dagger] = \bar{h}_{\alpha b} c_\alpha^\dagger + \frac{1}{4}\left(V^{\alpha\beta}_{\gamma b} - V^{\alpha\beta}_{b\gamma}\right)\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma - c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) \tag{5}$$

$$[H, c_a] = -\bar{h}_{a\alpha} c_\alpha + \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\left(c_\alpha^\dagger c_\beta c_\gamma - c_\gamma c_\beta c_\alpha^\dagger\right), \tag{6}$$

where we have hidden the summations on repeated indices using Einstein's notation.

Commutators calculations

First we calculate the commutator with $c_b^\dagger$ making use of the following commutator properties. We obtain

$$\left[c_\alpha^\dagger c_\beta, c_b^\dagger\right] = c_\alpha^\dagger \left\{c_\beta, c_b^\dagger\right\} - \cancel{\left\{c_b^\dagger, c_\alpha^\dagger\right\}} c_\beta = c_\alpha^\dagger \delta_{\beta b}$$
$$\left[c_\beta c_\alpha^\dagger, c_b^\dagger\right] = c_\beta \cancel{\left\{c_\alpha^\dagger, c_b^\dagger\right\}} - \left\{c_b^\dagger, c_\beta\right\} c_\alpha^\dagger = -c_\alpha^\dagger \delta_{\beta b}$$

and that

$$\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_b^\dagger\right] = c_\alpha^\dagger c_\beta^\dagger \left(c_\gamma \left\{c_\delta, c_b^\dagger\right\} - \left\{c_b^\dagger, c_\gamma\right\} c_\delta\right) + \left(c_\alpha^\dagger \cancel{\left\{c_\beta, c_b^\dagger\right\}} - \cancel{\left\{c_b^\dagger, c_\alpha^\dagger\right\}} c_\beta^\dagger\right) c_\gamma c_\delta = c_\alpha^\dagger c_\beta^\dagger c_\gamma \delta_{\delta b} - c_\alpha^\dagger c_\beta^\dagger c_\delta \delta_{\gamma b}$$
$$\left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_b^\dagger\right] = c_\delta c_\gamma \left(c_\beta^\dagger \cancel{\left\{c_\alpha^\dagger, c_b^\dagger\right\}} - \cancel{\left\{c_b^\dagger, c_\beta^\dagger\right\}} c_\alpha^\dagger\right) + \left(c_\delta \left\{c_\gamma, c_b^\dagger\right\} - \left\{c_b^\dagger, c_\delta\right\} c_\gamma\right) c_\beta^\dagger c_\alpha^\dagger = c_\delta c_\beta^\dagger c_\alpha^\dagger \delta_{\gamma b} - c_\gamma c_\beta^\dagger c_\alpha^\dagger \delta_{\delta b}$$

The complete commutator then reads

$$\begin{aligned}
[H, c_b^\dagger] &= \left[\frac{1}{2}\bar{h}_{\alpha\beta}\left(c_\alpha^\dagger c_\beta - c_\beta c_\alpha^\dagger\right) + \frac{1}{4}V^{\alpha\beta}_{\gamma\delta}\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta + c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger\right) + \frac{1}{2}\cancel{\tilde{h}_0}, c_b^\dagger\right] \\
&= \frac{1}{2}\bar{h}_{\alpha\beta}\left(\left[c_\alpha^\dagger c_\beta, c_b^\dagger\right] - \left[c_\beta c_\alpha^\dagger, c_b^\dagger\right]\right) + \frac{1}{4}V^{\alpha\beta}_{\gamma\delta}\left(\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_b^\dagger\right] + \left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_b^\dagger\right]\right) \\
&= \bar{h}_{\alpha\beta} c_\alpha^\dagger \delta_{\beta b} + \frac{1}{4}\left(V^{\alpha\beta}_{\gamma\delta} c_\alpha^\dagger c_\beta^\dagger c_\gamma \delta_{\delta b} - V^{\alpha\beta}_{\gamma\delta} c_\alpha^\dagger c_\beta^\dagger c_\delta \delta_{\gamma b} + V^{\alpha\beta}_{\gamma\delta} c_\delta c_\beta^\dagger c_\alpha^\dagger \delta_{\gamma b} - V^{\alpha\beta}_{\gamma\delta} c_\gamma c_\beta^\dagger c_\alpha^\dagger \delta_{\delta b}\right) \\
&= \bar{h}_{\alpha b} c_\alpha^\dagger + \frac{1}{4}\left(V^{\alpha\beta}_{\gamma b} - V^{\alpha\beta}_{b\gamma}\right)\left(c_\alpha^\dagger c_\beta^\dagger c_\gamma - c_\gamma c_\beta^\dagger c_\alpha^\dagger\right)
\end{aligned}$$

Analogous for the $[H, c_a]$ commutator, we have

$$\left[c_\alpha^\dagger c_\beta, c_a\right] = c_\alpha^\dagger \cancel{\{c_\beta, c_a\}} - \left\{c_a, c_\alpha^\dagger\right\} c_\beta = -c_\beta \delta_{a\alpha}$$
$$\left[c_\beta c_\alpha^\dagger, c_a\right] = c_\beta \left\{c_\alpha^\dagger, c_a\right\} - \cancel{\{c_a, c_\beta\}} c_\alpha^\dagger = c_\beta \delta_{a\alpha}$$

and that

$$\left[c_\alpha^\dagger c_\beta^\dagger c_\gamma c_\delta, c_a\right] = c_\alpha^\dagger c_\beta^\dagger \left(c_\gamma \cancel{\{c_\delta, c_a\}} - \cancel{\{c_a, c_\gamma\}} c_\delta\right) + \left(c_\alpha^\dagger \left\{c_\beta^\dagger, c_a\right\} - \left\{c_a, c_\alpha^\dagger\right\} c_\beta^\dagger\right) c_\gamma c_\delta = c_\alpha^\dagger c_\gamma c_\delta \delta_{\beta a} - c_\beta^\dagger c_\gamma c_\delta \delta_{a\alpha}$$
$$\left[c_\delta c_\gamma c_\beta^\dagger c_\alpha^\dagger, c_a\right] = c_\delta c_\gamma \left(c_\beta^\dagger \left\{c_\alpha^\dagger, c_a\right\} - \left\{c_a, c_\beta^\dagger\right\} c_\alpha^\dagger\right) + \left(c_\delta \cancel{\{c_\gamma, c_a\}} - \cancel{\{c_a, c_\delta\}} c_\gamma\right) c_\beta^\dagger c_\alpha^\dagger = c_\delta c_\gamma c_\beta^\dagger \delta_{\alpha a} - c_\delta c_\gamma c_\alpha^\dagger \delta_{a\beta}$$

The complete commutator then reads

$$
\begin{aligned}
[H, c_a] &= \left[\frac{1}{2}\bar{h}_{\alpha\beta}\left(c^\dagger_\alpha c_\beta - c_\beta c^\dagger_\alpha\right) + \frac{1}{4}V^{\alpha\beta}_{\gamma\delta}\left(c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta + c_\delta c_\gamma c^\dagger_\beta c^\dagger_\alpha\right) + \frac{1}{2}\tilde{\not{h}}_0, c_a\right] \\
&= \frac{1}{2}\bar{h}_{\alpha\beta}\left(\left[c^\dagger_\alpha c_\beta, c_a\right] - \left[c_\beta c^\dagger_\alpha, c_a\right]\right) + \frac{1}{4}V^{\alpha\beta}_{\gamma\delta}\left(\left[c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta, c_a\right] + \left[c_\delta c_\gamma c^\dagger_\beta c^\dagger_\alpha, c_a\right]\right) \\
&= -\bar{h}_{\alpha\beta}c_\beta\delta_{a\alpha} + \frac{1}{4}\left(V^{\alpha\beta}_{\gamma\delta}c^\dagger_\alpha c_\gamma c_\delta\delta_{\beta a} - V^{\alpha\beta}_{\gamma\delta}c^\dagger_\beta c_\gamma c_\delta\delta_{a\alpha} + V^{\alpha\beta}_{\gamma\delta}c_\delta c_\gamma c^\dagger_\beta\delta_{\alpha a} - V^{\alpha\beta}_{\gamma\delta}c_\delta c_\gamma c^\dagger_\alpha\delta_{a\beta}\right) \\
&= -\bar{h}_{a\beta}c_\beta + \frac{1}{4}\left(V^{\alpha a}_{\gamma\delta}c^\dagger_\alpha c_\gamma c_\delta - V^{a\beta}_{\gamma\delta}c^\dagger_\beta c_\gamma c_\delta + V^{a\beta}_{\gamma\delta}c_\delta c_\gamma c^\dagger_\beta - V^{\alpha a}_{\gamma\delta}c_\delta c_\gamma c^\dagger_\alpha\right) \\
&= -\bar{h}_{a\alpha}c_\alpha + \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\left(c^\dagger_\alpha c_\beta c_\gamma - c_\gamma c_\beta c^\dagger_\alpha\right)
\end{aligned}
$$

$$
\begin{aligned}
&\{c_a, c^\dagger_b\} = \{c^\dagger_b, c_a\} = \delta_{ab} \\
&\{c_a, c_b\} = \{c^\dagger_b, c^\dagger_a\} = 0
\end{aligned}
$$

$$
\begin{aligned}
&[AB, Z] = A\{B, Z\} - \{Z, A\}B \\
&[AB, Z] = A[B, Z] - [Z, A]B \\
&[AB, CD] = -A\left(C\{D, B\} - \{B, C\}D\right) - \left(C\{D, A\} - \{A, C\}D\right)B \\
&[ABCD, Z] = AB\left(C\{D, Z\} - \{Z, C\}D\right) + \left(A\{B, Z\} - \{Z, A\}B\right)CD
\end{aligned}
$$

Substituting Eqs.(5) and (6) back into Eq.(4) yields

$$
\begin{aligned}
-i\hbar\frac{d}{dt}\rho^{\mathrm{ee}}_{ab} &= \rho^{\mathrm{ee}}_{a\alpha}\bar{h}_{\alpha b} - \bar{h}_{a\alpha}\rho^{\mathrm{ee}}_{\alpha b} \\
&+ \frac{1}{4}\left(V^{\alpha\beta}_{\gamma b} - V^{\alpha\beta}_{b\gamma}\right)\left(\left\langle c^\dagger_\alpha c^\dagger_\beta c_\gamma c_a\right\rangle - \left\langle c_\gamma c^\dagger_\beta c^\dagger_\alpha c_a\right\rangle\right) \\
&+ \frac{1}{4}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\left(\left\langle c^\dagger_b c^\dagger_\alpha c_\beta c_\gamma\right\rangle - \left\langle c^\dagger_b c_\gamma c_\beta c^\dagger_\alpha\right\rangle\right)
\end{aligned} \tag{7}
$$

where we came back to the definition of the normal rDM in Eq.(**??**) for the first two terms. Notice, however, that the remaining interaction terms correspond to the expectation values of four-operators. For this, we introduce a mean-field approximation where we assume the two-particle expectation value to simply behave as a product of two one-particle expectation values. From this mean-field decoupling we can then make use of *Wick's theorem* as done in the Appendix. We obtain

$$
\left\langle c^\dagger_\alpha c^\dagger_\beta c_\gamma c_\delta\right\rangle \approx \rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{\delta\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{hh}}_{\delta\beta} + \rho^{\mathrm{ee}}_{\delta\alpha}\rho^{\mathrm{ee}}_{\gamma\beta} \tag{8}
$$

$$
\left\langle c_\alpha c^\dagger_\beta c^\dagger_\gamma c_\delta\right\rangle \approx \rho^{\mathrm{hh}}_{\beta\alpha}\rho^{\mathrm{ee}}_{\delta\gamma} - \rho^{\mathrm{hh}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{\delta\beta} + \rho^{\mathrm{eh}}_{\delta\alpha}\rho^{\mathrm{he}}_{\gamma\beta} \tag{9}
$$

$$
\left\langle c^\dagger_\alpha c_\beta c_\gamma c^\dagger_\delta\right\rangle \approx \rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{ee}}_{\delta\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{hh}}_{\delta\beta} + \rho^{\mathrm{he}}_{\delta\alpha}\rho^{\mathrm{eh}}_{\gamma\beta} \tag{10}
$$

As derived below in the gray appendices, the purely electronic rDM equation of motion can then be written in a time-dependent Hartree-Fock in a more compact and elegant manner as

$$
i\hbar\frac{d}{dt}\rho^{\mathrm{ee}}_{ab} = \left[h^{\mathrm{HF}}_{ab}, \rho^{\mathrm{ee}}_{ab}\right] - \rho^{\mathrm{ee}}_{ab}\left(\Sigma^{\Delta}_{ba}\right)^* + \Sigma^{\Delta}_{ab}\rho^{\mathrm{he}}_{ab} \tag{11}
$$

with the Hartree-Fock Hamiltonian being

$$
h^{HF}_{ab}(\rho) = h_{ab} + \Sigma^{H}_{ab}(\rho) + \Sigma^{F}_{ab}(\rho), \tag{12}
$$

identifying the Hartree, Fock and pairing self-energies (themselves dependent on the rDM) respectively as

$$
\begin{aligned}
\Sigma^{H}_{ab} &= \textstyle\sum_{\gamma\delta} V^{\delta a}_{b\gamma}\rho^{\mathrm{ee}}_{\gamma\delta} \\
\Sigma^{F}_{ab} &= -\textstyle\sum_{\gamma\delta} V^{a\delta}_{b\gamma}\rho^{\mathrm{ee}}_{\gamma\delta}
\end{aligned}
\quad \text{and} \quad
\Sigma^{\Delta}_{ab} = \sum_{\gamma\delta} V^{ab}_{\delta\gamma}\rho^{\mathrm{eh}}_{\gamma\delta} \tag{13}
$$

As a side note concerning excitons, notice that the Fock self-energy term in is an attractive interaction (due the negative sign) such that it lower the total energy of the system enabling the formation of the electron-hole bound states. Conversely, the Hartree self-energy term will control details of the excitation spectrum, such as spin-splitting. Also, note the symmetry $\Sigma^{\Delta}_{ab} = -\Sigma^{\Delta}_{ba}$.

Analogously for the anomalous EoM we have that

$$\frac{d}{dt}\rho^{\mathrm{eh}}_{ab} = \left\langle \frac{i}{\hbar}[H, c_b]c_a \right\rangle - \left\langle c_b \frac{i}{\hbar}[H, c_a] \right\rangle . \tag{14}$$

but instead perform the mean-field decoupling

$$\left\langle c^{\dagger}_{\alpha} c_{\beta} c_{\gamma} c_{\delta} \right\rangle \approx \rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{\delta\gamma} - \rho^{\mathrm{eh}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{\delta\beta} + \rho^{\mathrm{ee}}_{\delta\alpha}\rho^{\mathrm{eh}}_{\gamma\beta}, \tag{15}$$

$$\left\langle c_{\alpha} c^{\dagger}_{\beta} c_{\gamma} c_{\delta} \right\rangle \approx \rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{\delta\gamma} - \rho^{\mathrm{eh}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{\delta\beta} + \rho^{\mathrm{eh}}_{\delta\alpha}\rho^{\mathrm{ee}}_{\gamma\beta}. \tag{16}$$

$$\left\langle c_{\alpha} c_{\beta} c^{\dagger}_{\gamma} c_{\delta} \right\rangle \approx \rho^{\mathrm{eh}}_{\beta\alpha}\rho^{\mathrm{ee}}_{\delta\gamma} - \rho^{\mathrm{hh}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{\delta\beta} + \rho^{\mathrm{eh}}_{\delta\alpha}\rho^{\mathrm{hh}}_{\gamma\beta} \tag{17}$$

$$\left\langle c_{\alpha} c_{\beta} c_{\gamma} c^{\dagger}_{\delta} \right\rangle \approx \rho^{\mathrm{eh}}_{\beta\alpha}\rho^{\mathrm{hh}}_{\delta\gamma} - \rho^{\mathrm{eh}}_{\gamma\alpha}\rho^{\mathrm{hh}}_{\delta\beta} + \rho^{\mathrm{hh}}_{\delta\alpha}\rho^{\mathrm{eh}}_{\gamma\beta} \tag{18}$$

which yields the anomalous equation of motion

$$i\hbar\frac{d}{dt}\rho^{\mathrm{eh}}_{ab} = h^{\mathrm{HF}}_{ab}\,\rho^{\mathrm{eh}}_{ab} - \rho^{\mathrm{eh}}_{ab}h^{\mathrm{HF}}_{ba} + \Sigma^{\Delta}_{ab}\rho^{\mathrm{hh}}_{ab} - \rho^{\mathrm{ee}}_{ab}\Sigma^{\Delta}_{ab} \tag{19}$$

rDM EoM calculations

$$[H, c^{\dagger}_b] = h_{\alpha b}c^{\dagger}_{\alpha} + \frac{1}{2}(V^{\alpha\beta}_{\gamma b} - V^{\alpha\beta}_{b\gamma})c^{\dagger}_{\alpha}c^{\dagger}_{\beta}c_{\gamma}$$
$$[H, c_a] = -h_{a\alpha}c_{\alpha} + \frac{1}{2}(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma})c^{\dagger}_{\alpha}c_{\beta}c_{\gamma}$$

Using the mean-field decoupling in Eq.(**??**), the *normal* EoM interaction terms becomes

$$-\frac{1}{2}(V^{\alpha\beta}_{\gamma b} - V^{\alpha\beta}_{b\gamma})(\rho^{\mathrm{he}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{a\beta} + \rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{ee}}_{\gamma\beta})$$
$$-\frac{1}{2}(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma})(\rho^{\mathrm{he}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta} - \rho^{\mathrm{ee}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha} + \rho^{\mathrm{ee}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha})$$

with the purely electronic reading

$$-\rho^{\mathrm{ee}}_{a\beta}(V^{\alpha\beta}_{b\gamma} - V^{\beta\alpha}_{b\gamma})\rho^{\mathrm{ee}}_{\gamma\alpha} + (V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma})\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{ee}}_{\beta b}$$
$$= -\rho^{\mathrm{ee}}_{a\beta}\Sigma^{\mathrm{HF}}_{\beta b} + \Sigma^{\mathrm{HF}}_{a\beta}\,\rho^{\mathrm{ee}}_{\beta b} = [\Sigma_{\mathrm{HF}}, \rho_{\mathrm{ee}}]|_{ab}$$

(no index is a matrix since I reserve the bold specifically to denote the spin structure) and the anomalous terms reads

$$-\frac{1}{2}\rho^{\mathrm{eh}}_{a\gamma}(V^{\beta\alpha}_{b\gamma} - V^{\beta\alpha}_{\gamma b})(\rho^{\mathrm{eh}}_{\alpha\beta})^{*} - \frac{1}{2}(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma})\rho^{\mathrm{eh}}_{\gamma\beta}\rho^{\mathrm{he}}_{\alpha b}$$
$$= -\frac{1}{2}\rho^{\mathrm{eh}}_{a\gamma}(\Sigma^{\Delta}_{b\gamma} - \Sigma^{\Delta}_{\gamma b})^{*} - \frac{1}{2}(\Sigma^{\Delta}_{\alpha a} - \Sigma^{\Delta}_{a\alpha})\rho^{\mathrm{he}}_{\alpha b}$$
$$= \rho_{\mathrm{eh}} \cdot \Sigma^{*}_{\Delta}|_{ab} + \Sigma_{\Delta} \cdot \rho_{\mathrm{he}}|_{ab}$$

The non-interacting terms is trivially $-[h, \rho_{\mathrm{ee}}]_{ab}$. Analogously for the *anomalous* EoM, we have trivially non-interacting term $-[\rho_{\mathrm{eh}} \cdot h^{T}]_{ab} - [h \cdot \rho_{\mathrm{eh}}]_{ab}$. For the interaction terms we perform instead the mean-field decoupling of $\langle c^{\dagger}_{\alpha}c_{\beta}c_{\gamma}c_{a}\rangle$ and $\langle c_b c^{\dagger}_{\alpha}c_{\beta}c_{\gamma}\rangle$, we obtain

$$-\frac{1}{2}(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma})(\rho^{\mathrm{ee}}_{\beta\alpha}\rho^{\mathrm{eh}}_{a\gamma} - \rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{a\beta} + \rho^{\mathrm{ee}}_{a\alpha}\rho^{\mathrm{eh}}_{\gamma\beta})$$
$$-\frac{1}{2}(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma})(\rho^{\mathrm{hh}}_{\alpha b}\rho^{\mathrm{eh}}_{\gamma\beta} - \rho^{\mathrm{eh}}_{\beta b}\rho^{\mathrm{ee}}_{\gamma\alpha} + \rho^{\mathrm{eh}}_{\gamma b}\rho^{\mathrm{ee}}_{\beta\alpha})$$

The first term reading

$$
\begin{aligned}
&\rho^{\mathrm{eh}}_{a\gamma}(V^{\alpha b}_{\gamma\beta} - V^{b\alpha}_{\gamma\beta})\rho^{\mathrm{ee}}_{\beta\alpha} - \frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}(V^{\alpha b}_{\beta\gamma} - V^{b\alpha}_{\beta\gamma})\rho^{\mathrm{eh}}_{\gamma\beta} \\
=&\rho^{\mathrm{eh}}_{a\gamma}\Sigma^{\mathrm{HF}}_{b\gamma} - \frac{1}{2}\rho^{\mathrm{ee}}_{a\alpha}(\Sigma^{\Delta}_{\alpha b} - \Sigma^{\Delta}_{b\alpha}) \\
=&\ \rho_{\mathrm{eh}} \cdot \Sigma^{*}_{\mathrm{HF}}\big|_{ab} - \rho_{\mathrm{ee}} \cdot \Sigma_{\Delta}\big|_{ab}
\end{aligned}
$$

and the second one reading

$$
\begin{aligned}
&-\frac{1}{2}\left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{eh}}_{\gamma\beta}\rho^{\mathrm{hh}}_{\alpha b} + \left(V^{\alpha a}_{\beta\gamma} - V^{a\alpha}_{\beta\gamma}\right)\rho^{\mathrm{ee}}_{\gamma\alpha}\rho^{\mathrm{eh}}_{\beta b} \\
=&-\frac{1}{2}\left(\Sigma^{\Delta}_{\alpha a} - \Sigma^{\Delta}_{a\alpha}\right)\rho^{\mathrm{hh}}_{\alpha b} + \Sigma^{\mathrm{HF}}_{a\beta}\rho^{\mathrm{eh}}_{\beta b} \\
=&\ \Sigma_{\Delta} \cdot \rho_{\mathrm{hh}}\big|_{ab} + \Sigma_{\mathrm{HF}} \cdot \rho_{\mathrm{eh}}\big|_{ab}
\end{aligned}
$$

Confirming that the result actually checks out

$$
\begin{aligned}
&\begin{pmatrix} h^{\mathrm{HF}}_{ab} & \Sigma^{\Delta}_{ab} \\ \left(\Sigma^{\Delta}_{ba}\right)^{*} & -h^{\mathrm{HF}}_{ba} \end{pmatrix}\begin{pmatrix} \rho^{\mathrm{ee}}_{ab} & \rho^{\mathrm{eh}}_{ab} \\ \rho^{\mathrm{he}}_{ab} & \rho^{\mathrm{hh}}_{ab} \end{pmatrix} - \begin{pmatrix} \rho^{\mathrm{ee}}_{ab} & \rho^{\mathrm{eh}}_{ab} \\ \rho^{\mathrm{he}}_{ab} & \rho^{\mathrm{hh}}_{ab} \end{pmatrix}\begin{pmatrix} h^{\mathrm{HF}}_{ab} & \Sigma^{\Delta}_{ab} \\ \left(\Sigma^{\Delta}_{ba}\right)^{*} & -h^{\mathrm{HF}}_{ba} \end{pmatrix} \\
=&\begin{bmatrix} \left(h^{\mathrm{HF}}_{ab}\rho^{\mathrm{ee}}_{ab} + \Sigma^{\Delta}_{ab}\rho^{\mathrm{he}}_{ab}\right) & \left(h^{\mathrm{HF}}_{ab}\rho^{\mathrm{eh}}_{ab} + \Sigma^{\Delta}_{ab}\rho^{\mathrm{hh}}_{ab}\right) \\ \circ & \circ \end{bmatrix} - \begin{bmatrix} \left(\rho^{\mathrm{ee}}_{ab}h^{\mathrm{HF}}_{ab} - \rho^{\mathrm{eh}}_{ab}\left(\Sigma^{\Delta}_{ba}\right)^{*}\right) & \left(\rho^{\mathrm{ee}}_{ab}\Sigma^{\Delta}_{ab} - \rho^{\mathrm{eh}}_{ab}h^{\mathrm{HF}}_{ba}\right) \\ \circ & \circ \end{bmatrix} \\
=&\begin{bmatrix} \left(\left[h^{\mathrm{HF}}_{ab}, \rho^{\mathrm{ee}}_{ab}\right] + \Sigma^{\Delta}_{ab}\rho_{\mathrm{he}} + \rho^{\mathrm{eh}}_{ab}\left(\Sigma^{\Delta}_{ba}\right)^{*}\right) & \left(h^{\mathrm{HF}}_{ab}\rho^{\mathrm{eh}}_{ab} - \rho^{\mathrm{eh}}_{ab}h^{\mathrm{HF}}_{ba} - \rho^{\mathrm{ee}}_{ab}\Sigma^{\Delta}_{ab} + \Sigma^{\Delta}_{ab}\rho^{\mathrm{hh}}_{ab}\right) \\ \circ & \circ \end{bmatrix}
\end{aligned}
$$

$$
\begin{aligned}
&\rho^{\mathrm{hh}}_{ab} = \left\langle c_b c^{\dagger}_a \right\rangle = -\left\langle c^{\dagger}_a c_b \right\rangle + \delta_{ab} = -\rho^{\mathrm{ee}}_{ba} + \delta_{ab} \\
&\rho^{\mathrm{he}}_{ab} = \left\langle c^{\dagger}_b c^{\dagger}_a \right\rangle = \left\langle c_a c_b \right\rangle^{\dagger} = \left(\rho^{\mathrm{eh}}_{ba}\right)^{\dagger} \\
&\{c_a, c_b\} = 0 \Leftrightarrow c_a c_b = -c_b c_a \Rightarrow \rho^{\mathrm{eh}}_{ba} = -\rho^{\mathrm{eh}}_{ab} \\
&\left\{c^{\dagger}_a, c^{\dagger}_b\right\} = 0 \Leftrightarrow c^{\dagger}_a c^{\dagger}_b = -c^{\dagger}_b c^{\dagger}_a \Rightarrow \rho^{\mathrm{eh}}_{ba} = -\rho^{\mathrm{eh}}_{ab} \\
&\Sigma^{\Delta}_{ab} = V^{ab}_{\delta\gamma}\rho^{\mathrm{eh}}_{\gamma\delta} = V^{ba}_{\gamma\delta}\rho^{\mathrm{eh}}_{\gamma\delta} \overset{\delta \rightleftarrows \gamma}{=} V^{ba}_{\delta\gamma}\rho^{\mathrm{eh}}_{\delta\gamma} = V^{ba}_{\delta\gamma}\left(-\rho^{\mathrm{eh}}_{\gamma\delta}\right) = -\Sigma^{\Delta}_{ba}
\end{aligned}
$$

Putting these two results together yields a Nambu rDM EoM in terms of an effective Bogoliubov-de Gennes (BdG) Hamiltonian as simply as

$$
i\hbar\frac{d}{dt}\rho_{ab} = \left[H^{\mathrm{BdG}}_{ab}, \rho_{ab}\right] \ \text{ with } H_{\mathrm{BdG}} = \begin{pmatrix} h^{\mathrm{HF}}_{ab} & \Sigma^{\Delta}_{ab} \\ \left(\Sigma^{\Delta}_{ba}\right)^{*} & -h^{\mathrm{HF}}_{ba} \end{pmatrix} \tag{20}
$$

C. Spin-1/2 Hubbard model in the Wannier limit

Furthermore, consider now the approximation where the orbitals are ultra-localized (known as Wannier orbitals) such that the interaction potential, written in terms of an arbitrary electronic basis $\{\phi_\alpha(\mathbf{r})\}$, approximates to

$$
V^{\alpha\beta}_{\gamma\delta} = \int d\mathbf{r}d\mathbf{r}'\phi^{*}_{\alpha}(\mathbf{r})\phi^{*}_{\beta}(\mathbf{r}')V(\mathbf{r}-\mathbf{r}')\phi_{\gamma}(\mathbf{r}')\phi_{\delta}(\mathbf{r}) \approx V^{\alpha\beta}\delta_{\alpha\delta}\delta_{\beta\gamma} \tag{21}
$$

with $V^{\alpha\beta} \equiv V(\mathbf{r}_\alpha - \mathbf{r}_\beta)$. Within the scope of this approximation the interaction Hamiltonian will becomes simply $H_{\mathrm{int}} = \frac{1}{2}\sum_{\alpha\beta} V^{\alpha\beta}c^{\dagger}_{\alpha}c^{\dagger}_{\beta}c_{\beta}c_{\alpha}$. The Hartree and Fock self-energies become

$$
\begin{matrix} \Sigma^{H}_{ab} = \sum_{\gamma} V^{a\gamma}\rho^{\mathrm{ee}}_{\gamma\gamma}\delta_{ab} \\ \Sigma^{F}_{ab} = -V^{ab}\rho^{\mathrm{ee}}_{ab} \end{matrix} \quad \text{and} \quad \Sigma^{\Delta}_{ab} = V^{ab}\rho^{\mathrm{eh}}_{ab} \tag{22}
$$

Furthermore, for the sake of getting the problem in a tight-binding scheme, we isolate the degrees of freedom into purely spatial and spin indices $a \to i\sigma_i$ such that

$$\boldsymbol{\rho}_{ij} = \begin{pmatrix} \boldsymbol{\rho}_{ij}^{\mathrm{ee}} & \boldsymbol{\rho}_{ij}^{\mathrm{eh}} \\ \boldsymbol{\rho}_{ji}^{\mathrm{he}} & \boldsymbol{\rho}_{ji}^{\mathrm{hh}} \end{pmatrix} = \left(\begin{array}{cc|cc} \rho_{i\uparrow j\uparrow}^{\mathrm{ee}} & \rho_{i\uparrow j\downarrow}^{\mathrm{ee}} & \rho_{i\uparrow j\uparrow}^{\mathrm{eh}} & \rho_{i\uparrow j\downarrow}^{\mathrm{eh}} \\ \rho_{i\downarrow j\uparrow}^{\mathrm{ee}} & \rho_{i\downarrow j\downarrow}^{\mathrm{ee}} & \rho_{i\downarrow j\uparrow}^{\mathrm{eh}} & \rho_{i\downarrow j\downarrow}^{\mathrm{eh}} \\ \hline \rho_{i\uparrow j\uparrow}^{\mathrm{he}} & \rho_{i\uparrow j\downarrow}^{\mathrm{he}} & \rho_{i\uparrow j\uparrow}^{\mathrm{hh}} & \rho_{i\downarrow j\uparrow}^{\mathrm{hh}} \\ \rho_{i\downarrow j\uparrow}^{\mathrm{he}} & \rho_{i\downarrow j\downarrow}^{\mathrm{he}} & \rho_{i\uparrow j\downarrow}^{\mathrm{hh}} & \rho_{i\downarrow j\downarrow}^{\mathrm{hh}} \end{array}\right) \tag{23}$$

Note the use of the bold to denote that an object possesses a spin structure. Moreover, notice that, since we considered the interaction to be spin-independent, we also have that $V^{i\sigma_i j\sigma_j} \equiv V^{ij}$. In this basis, the self-energies read as

$$\begin{aligned} \Sigma^{H}_{i\sigma_i j\sigma_j} &= \sum_{\ell} V^{i\ell} \left(\rho_{\ell\uparrow\ell\uparrow}^{\mathrm{ee}} + \rho_{\ell\downarrow\ell\downarrow}^{\mathrm{ee}}\right) \delta_{ij}\delta_{\sigma_i\sigma_j}, \\ \Sigma^{F}_{i\sigma_i j\sigma_j} &= -V^{ij}\rho_{i\sigma_i j\sigma_j}^{\mathrm{ee}} \text{ and } \Sigma^{\Delta}_{i\sigma_i j\sigma_j} = V^{ij}\rho_{i\sigma_i j\sigma_j}^{\mathrm{eh}}, \end{aligned} \tag{24}$$

with ℓ a dummy site index. Notice that the Hartree term is exclusively an onsite due to the δ_{ij}. If we now define a new object related to the Nambu rDM but specifically having the same symmetries as the Nambu Hamiltonian, this being

$$\tilde{\boldsymbol{\rho}}_{ij} = \begin{pmatrix} \boldsymbol{\rho}_{ij}^{\mathrm{ee}} & -\boldsymbol{\rho}_{ij}^{\mathrm{eh}} \\ -\left(\boldsymbol{\rho}_{ji}^{\mathrm{eh}}\right)^{*} & -\boldsymbol{\rho}_{ji}^{\mathrm{ee}} \end{pmatrix}, \tag{25}$$

we can then write the total self-energy of the system in an elegant and compact form as

$$\boldsymbol{\Sigma}_{ij} = \frac{1}{2}\sum_{\ell} V^{i\ell}\mathrm{Tr}\left(\left[\tau_z \otimes \sigma_0\right]\tilde{\boldsymbol{\rho}}_{\ell\ell}\right)\left[\tau_z \otimes \sigma_0\right]\delta_{ij} - V^{ij}\tilde{\boldsymbol{\rho}}_{ij}, \tag{26}$$

with both σ and τ as Pauli matrices but operating in their respectively spin and electron-hole subspaces the Hamiltonian. In the limiting case of a spin-1/2 Hubbard model one considers $V^{ij} = U\delta_{ij}$ such that mean field decoupling gives rise to just the onsite term,

$$\boldsymbol{\Sigma}_{ij} = \frac{1}{2}U\left[\mathrm{Tr}\left(\left[\tau_z \otimes \sigma_0\right]\tilde{\boldsymbol{\rho}}_{ii}\right)\left[\tau_z \otimes \sigma_0\right] - 2\tilde{\boldsymbol{\rho}}_{ij}\right]\delta_{ij}. \tag{27}$$

Hubbard self-energy

$$\begin{aligned} \boldsymbol{\Sigma}_i &= \begin{pmatrix} \sum_{\ell} V^{i\ell}\left(\rho_{\ell\uparrow\ell\uparrow}^{\mathrm{ee}} + \rho_{\ell\downarrow\ell\downarrow}^{\mathrm{ee}}\right)\mathbb{1} - V^{ij}\boldsymbol{\rho}_{ij}^{\mathrm{ee}} & V^{ij}\boldsymbol{\rho}_{ij}^{\mathrm{eh}} \\ \left[V^{ij}\boldsymbol{\rho}_{ij}^{\mathrm{eh}}\right]^{\dagger} & -\left[\sum_{\ell} V^{i\ell}\left(\rho_{\ell\uparrow\ell\uparrow}^{\mathrm{ee}} + \rho_{\ell\downarrow\ell\downarrow}^{\mathrm{ee}}\right)\mathbb{1} - V^{ij}\boldsymbol{\rho}_{ij}^{\mathrm{ee}}\right]^{T} \end{pmatrix} \\ &= \frac{1}{2}\sum_{\ell} V^{i\ell}\mathrm{Tr}\left(\tau_z\tilde{\boldsymbol{\rho}}_{\ell\ell}\right)\begin{pmatrix} \mathbb{1} & \\ & -\mathbb{1} \end{pmatrix} - V^{ij}\begin{pmatrix} \boldsymbol{\rho}_{ij}^{\mathrm{ee}} & -\boldsymbol{\rho}_{ij}^{\mathrm{eh}} \\ -\left(\boldsymbol{\rho}_{ji}^{\mathrm{eh}}\right)^{*} & -\boldsymbol{\rho}_{ji}^{\mathrm{ee}} \end{pmatrix} \\ &= \frac{1}{2}\sum_{\ell} V^{i\ell}\mathrm{Tr}\left(\tau_z\tilde{\boldsymbol{\rho}}_{\ell\ell}\right)\left[\tau_z \otimes \sigma_0\right] - V^{ij}\tilde{\boldsymbol{\rho}}_{ij} \end{aligned}$$

where the trace reads as

$$\mathrm{Tr}\left(\tau_z\tilde{\boldsymbol{\rho}}_{\ell}\right) = \mathrm{Tr}\left(\left(\begin{array}{cc|cc} \rho_{\ell\uparrow\ell\uparrow}^{\mathrm{ee}} & \rho_{\ell\uparrow\ell\downarrow}^{\mathrm{ee}} & -\rho_{\ell\uparrow\ell\uparrow}^{\mathrm{eh}} & -\rho_{\ell\uparrow\ell\downarrow}^{\mathrm{eh}} \\ \rho_{\ell\downarrow\ell\uparrow}^{\mathrm{ee}} & \rho_{\ell\downarrow\ell\downarrow}^{\mathrm{ee}} & -\rho_{\ell\downarrow\ell\uparrow}^{\mathrm{eh}} & -\rho_{\ell\downarrow\ell\downarrow}^{\mathrm{eh}} \\ \hline -\left[\rho_{\ell\uparrow\ell\uparrow}^{\mathrm{he}}\right] & -\left[\rho_{\ell\uparrow\ell\downarrow}^{\mathrm{he}}\right] & -\left[-\rho_{\ell\uparrow\ell\uparrow}^{\mathrm{ee}}\right] & -\left[\rho_{\ell\downarrow\ell\uparrow}^{\mathrm{ee}}\right] \\ -\left[\rho_{\ell\downarrow\ell\uparrow}^{\mathrm{he}}\right] & -\left[\rho_{\ell\downarrow\ell\downarrow}^{\mathrm{he}}\right] & -\left[\rho_{\ell\uparrow\ell\downarrow}^{\mathrm{ee}}\right] & -\left[-\rho_{\ell\downarrow\ell\downarrow}^{\mathrm{ee}}\right] \end{array}\right)\right) = 2\left(\rho_{\ell\uparrow\ell\uparrow}^{\mathrm{ee}} + \rho_{\ell\downarrow\ell\downarrow}^{\mathrm{ee}}\right)$$

II. GAP EQUATION

The typical textbook gap equation [???] describes the self-consistent pairing Δ of infinite system in a Hubbard scheme, reading

$$\Delta = -\frac{U}{N}\sum_k \frac{\Delta}{2E_k}\tanh\left(\frac{E_k}{2k_{\mathrm{B}}T}\right), \tag{28}$$

with U the onsite Hubbard potential, E_k the k-state eigen-energies and k_B and T the Boltzmann constant and temperature.

The main objective of this section is to derive the BCS gap equation from our Hartree-Fock-Bogoliubov approach with the additional generalization of allowing for a Zeeman field and spin orbit coupling, accounting not only for the self-consistent pairing but also the self-consistently induced magnetization and doping.

A. Without SOC

Consider the non-interacting Hamiltonian H_0 for an infinite nanowire in the presence of an external Zeeman field V_{Z},

$$H_0(k) = \sum_s c^\dagger_{ks}\left[(\epsilon_k - \mu)\delta_{ss'} + V_Z\sigma_z\right]c_{ks'}, \tag{29}$$

with $c^\dagger(c)$ fermionic creation (annihilation) operators specified by its spin degree of freedom $s = \uparrow / \downarrow$ and momentum k, and ϵ_k the respective spin independent (hence the $\delta_{ss'}$) energy and μ the chemical potential. Once more, both σ and τ are Pauli matrices operating in their respectively spin and electron-hole subspaces the Hamiltonian.

One can reparametrizaze the onsite Hartree-Fock-Bogoliubov self-energy in the Wannier-Hubbard limit in Eq.(27) with the symmetries of Eq.(2) in terms of the self-consistently induced magnetization δV_{Z}, doping $\delta\mu$ and pairing Δ (assumed to be real) as

$$\mathbf{\Sigma}_{ii} = \left(\delta V_{\mathrm{Z}}\left[\tau_z\otimes\sigma_z\right] - \delta\mu\left[\tau_z\otimes\sigma_0\right]\right) - \Delta\left[\tau_y\otimes i\sigma_y\right]. \tag{30}$$

Comparing this form with Eq.(2), we can identify the diagonal and off-diagonal block respectively as

$$-\Delta i\sigma_y = U\begin{pmatrix} 0 & \rho^{\mathrm{eh}}_{i\uparrow i\downarrow} \\ \rho^{\mathrm{eh}}_{i\downarrow i\uparrow} & 0 \end{pmatrix} \quad \text{and} \quad \delta V_{\mathrm{Z}}\sigma_z - \delta\mu\sigma_0 = \frac{1}{2}U\begin{pmatrix} \rho^{\mathrm{ee}}_{i\downarrow i\downarrow} & 0 \\ 0 & \rho^{\mathrm{ee}}_{i\uparrow i\uparrow} \end{pmatrix} \tag{31}$$

Let us first focus on the pairing term and then on the normal one. Given that $\rho^{\mathrm{eh}}_{i\uparrow j\downarrow} = -\rho^{\mathrm{eh}}_{i\downarrow j\uparrow}$ the pairing term becomes even simpler reading $\Delta = U\rho^{\mathrm{eh}}_{i\uparrow i\downarrow}$. The superconductive correlation $\rho^{\mathrm{eh}}_{i\uparrow i\downarrow}$, which is being described in orbital space, can then be Fourier transformed into the momentum space as

$$\boldsymbol{\rho}_{ij} = \frac{1}{N}\sum_k \langle\check{\boldsymbol{c}}^\dagger_k\otimes\check{\boldsymbol{c}}_k\rangle e^{ik(x_j - x_i)} \tag{32}$$

with k summed over the Brillouin zone and N the total number of sites in the system. Furthermore, the fermionic operators $c^\dagger_{k\uparrow}/c_{k\downarrow}$ can then be represented in terms of the eigenvalues $E_{k\pm}$ and (normalized) eigenvectors $(u_{k\pm}, v_{k\pm})$ of the $H_0(k)$ Hamiltonian in Eq.(29), namely

$$H^{\mathrm{e}\uparrow/h\downarrow}(k) = \begin{pmatrix} \tilde{\epsilon}_k + V_{\mathrm{Z}} & \Delta \\ \Delta & -\tilde{\epsilon}_k + V_{\mathrm{Z}} \end{pmatrix}$$

Hence, the superconductive correlation reads as

$$\rho^{\mathrm{eh}}_{i\uparrow i\downarrow} = \frac{1}{N}\sum_k\left[u^*_{k+}v_{k+}f(E_k) + u^*_{k-}v_{k-}f(E_{k-})\right]$$

where f is the Fermi-Dirac distribution.

B. Zero temperature approximations

According to Pablo's calculation, the self-consistent gap equation with Zeeman reads as

$$\Delta = -U \sum_k \frac{\Delta}{4\tilde{E}_k} \left[\tanh\left(\frac{\tilde{V}_\mathrm{Z} + \tilde{E}_k}{2k_B T} \right) - \tanh\left(\frac{\tilde{V}_\mathrm{Z} - \tilde{E}_k}{2k_B T} \right) \right]$$

where the tilded quantities are dressed, already accounting for the U-induced shifts, i.e $\tilde{\mu} = \mu - \delta\mu$ and $\tilde{V}_\mathrm{Z} = V_\mathrm{Z} + \delta V_\mathrm{Z}$, and E_k the eigenenergies given by $\tilde{E}_k = \sqrt{\Delta^2 + \tilde{\epsilon}_k^2}$ with $\tilde{\epsilon}_k = \epsilon_k - \tilde{\mu}$. Furthermore, the self-consistent equation for the induced Zeeman δV_Z and chemical potential $\delta\mu$ are

$$\begin{aligned}
\delta\mu =& \frac{U}{4} + \frac{U}{4N} \sum_k \frac{\tilde{\epsilon}_k}{2\tilde{E}_k} \left[\tanh\left(\frac{\tilde{V}_\mathrm{Z} + \tilde{E}_k}{2k_B T} \right) - \tanh\left(\frac{\tilde{V}_\mathrm{Z} - \tilde{E}_k}{2k_B T} \right) \right] \\
\delta V_\mathrm{Z} =& -\frac{U}{2N} \sum_k \frac{\sinh\left(\frac{\tilde{V}_Z}{k_B T}\right)}{\cosh\left(\frac{\tilde{V}_Z}{k_B T}\right) + \cosh\left(\frac{\tilde{E}_k}{k_B T}\right)}
\end{aligned}$$

Within the zero temperature regime $T \to 0K$ one can roughly approximate

$$\begin{aligned}
\delta V_\mathrm{Z} =& -\frac{U}{2N} \sum_k \frac{\sinh\left(\frac{\tilde{V}_Z}{k_B T}\right)}{\cosh\left(\frac{\tilde{V}_Z}{k_B T}\right) + \cosh\left(\frac{\tilde{E}_k}{k_B T}\right)} \\
\approx& -\frac{U}{2N} \sum_k \frac{1}{2} \frac{\sinh(1/T)}{\cosh(1/T)} \\
=& -\frac{U}{2N} \sum_k \frac{1}{2} \tanh(1/T) \overset{T\to 0}{\longrightarrow} -\frac{U}{4N} \sum_k 1
\end{aligned}$$

Moreover, and noting that

$$\begin{aligned}
\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}} \\
\overset{\chi = e^{-2x}}{\longrightarrow} \frac{1-\chi}{1+\chi} \overset{x\to\infty}{\longrightarrow} 1 - 2\chi + \mathcal{O}(\chi^2)
\end{aligned}$$

and thus

$$\begin{aligned}
&\tanh(x+y) - \tanh(x-y) \\
=& \frac{\tanh(x) + \tanh(y)}{1 + \tanh(x)\tanh(y)} - \frac{\tanh(x) + \tanh(-y)}{1 + \tanh(x)\tanh(-y)} \\
\approx& \frac{(1-2\chi) + (1-2\xi)}{1 + (1-2\chi)(1-2\xi)} - \frac{(1-2\chi) - (1-2\xi)}{1 - (1-2\chi)(1-2\xi)} \\
\approx& 1 + \frac{\chi - \xi}{\chi + \xi} = 1 + \frac{1 - e^{2(x-y)}}{1 + e^{2(x-y)}} = 2\frac{1}{e^{2(x-y)} + 1},
\end{aligned}$$

the remaining self-consistent equation reads as

$$\begin{aligned}
\Delta \approx& -U \sum_k \frac{\Delta}{2\tilde{E}_k} f\left(\tilde{V}_\mathrm{Z} - \tilde{E}_k\right), \\
\delta\mu \approx& \frac{U}{4} + \frac{U}{4N} \sum_k \frac{\tilde{\epsilon}_k}{\tilde{E}_k} f\left(\tilde{V}_\mathrm{Z} - \tilde{E}_k\right).
\end{aligned}$$